



AMAZON SDE INTERVIEW EXPERIENCE

Amazon is one of the world's most influential technology and innovation companies, excelling in global e-commerce, cloud infrastructure (AWS), artificial intelligence, and consumer technology. As a core member of the MAANG group (Meta, Amazon, Apple, Netflix, Google), it continues to drive large-scale digital transformation and offers developers diverse, high-impact opportunities across cutting-edge domains.

ROLE OVERVIEW

- **Position:** Software Development Engineer (SDE) Intern
 - **Duration:** 6 Months (performance-based FTE conversion possible)
 - **Selection Process:** 2 Rounds (Online Assessment + Technical Interview)
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ROUND 1 – ONLINE ASSESSMENT

The online test was conducted on HackerRank and included two coding problems followed by personality-based questions assessing behavioral alignment and problem-solving attitude. The total duration was 70 minutes, and **optimal solutions** were required to pass hidden test cases.

1. Problem Solving

Question 1 – Dataset Cleaning Cost

You are given a string dataset of even length consisting of lowercase English letters. You can remove any two characters in one operation:

- If both are the same $\rightarrow$ cost = matchCost
- If different $\rightarrow$ cost = mismatchCost

Your goal is to remove all characters while minimizing the total cost.

Example:

- dataset = "ouio"
- matchCost = 2
- mismatchCost = 4

Step-by-step Explanation:

1. The string is "ouio" $\rightarrow$ characters: 'o', 'o', 'u', 'i'.
2. Remove 'o' and 'o' $\rightarrow$ cost = 2 $\rightarrow$ remaining string = "ui".
3. Remove 'u' and 'i' $\rightarrow$ cost = 4.

Total Cost = 2 + 4 = 6

Question 2 – System Stabilization

You are given a string pipeline representing service nodes. A node type failedService disrupts the system:

- Every second, each occurrence of failedService removes the preceding character.
- All removals happen simultaneously every second.
- Continue until no more removals occur. Return the number of seconds for stabilization.

Example:

- pipeline = "database"
- failedService = 'a'

Step-by-step Explanation:

- Initial pipeline: "database"
- Time = 1 → Each 'a' removes preceding characters → removes 'd' and 't' → remaining string = "aaase"
- Time = 2 → Each 'a' removes preceding characters → remaining string = "ase"

Output = 2 seconds

2. Workstyle Assessment

A short behavioral test based on **Amazon's Leadership Principles** like Customer Obsession, Ownership, and Deliver Results.

- Example statements:
 - "I take initiative even when tasks aren't assigned to me."
 - "I try to simplify complex problems."

Tips:

- Be consistent with answers.
 - Show ownership, teamwork, curiosity, and customer focus.
 - Avoid extremes like "Always Disagree."
 - Review Amazon's Leadership Principles before the test.
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3. Work Simulation Assessment

A situational judgment test simulating real-world scenarios faced by Amazon engineers.

- Example scenario:

"A teammate finds a critical bug close to deadline — what's your first step?"

Tips:

- Prioritize customer impact and data-driven choices.
- Show ownership and balance speed with quality.
- Collaborate and consider team input.

Both assessments are **screening rounds** — strong coding alone isn't enough; alignment with Leadership Principles is equally important.

After the coding section, candidates completed a Work Simulation and reasoning questions aligned with Amazon's Leadership Principles. **107 candidates** were shortlisted for the technical interview.

ROUND 2 – TECHNICAL INTERVIEW

- **Mode:** Online (Amazon Chime)
- **Duration:** 60 minutes (completed early – ~40 minutes)
- **Focus:** DSA problem-solving, optimal approach discussion, structured reasoning

The interview began with a brief introduction, followed by two live coding problems emphasizing **optimal solutions**.

Question 1 – Word Break

Determine if a string `s` can be segmented into words from `wordDict` (reuse allowed). Optimal DP/memoization solutions expected.

- Example:
 - o `s = "applepenapple"`
 - o `wordDict = ["apple", "pen"]`
 - o Output: true
 - o Explanation: "applepenapple" → "apple pen apple"

Question 2 – Sum of Prefix Scores of Strings

Given an array of words, the score of a string is the count of words that have it as a prefix. Efficient Trie-based logic expected.

- Example:
 - o `words = ["abc", "ab", "bc", "b"]`
 - o Output = [5, 4, 3, 2]
 - o Explanation:
 - "abc" → prefixes: a, ab, abc → counts: 2 + 2 + 1 = 5
 - "ab" → a, ab → 2 + 2 = 4
 - "bc" → b, bc → 2 + 1 = 3
 - "b" → b → 2

Bonus Discussion – MST Edge Classification

Discussion on Critical and Pseudo-Critical Edges in MST, Kruskal's Algorithm, Union-Find, and identifying edges that increase MST weight when removed.

Finally, I asked a few questions about **team culture, collaboration, and work-life balance** at Amazon, such as:

- How does the team encourage knowledge sharing and mentorship?
- What does a typical day look like for an SDE Intern?
- Opportunities for learning and contributing to impactful projects.

This led to a **positive and engaging closing conversation** with the interviewer.

RESULT

Pleased to have realized my dream — selected for the Amazon SDE Internship (6 months).

LEARNINGS & TIPS

- Focus on clarity of explanation — Amazon values reasoning as much as code.
 - Think aloud and explain your thought process clearly.
 - Revise Trees, Graphs, DP, and Heaps.
 - Understand Amazon's Leadership Principles.
 - Always aim for **optimal solutions** — brute force may fail hidden cases.
 - Be confident and ask thoughtful questions to the interviewer.
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MY LEARNING JOURNEY

- DSA Concepts: Striver (A–Z)
 - Writing Clean Code: NeetCode (YouTube)
 - Problem-Solving Mindset: Larry (YouTuber)
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LET'S CONNECT

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